Mol Gen Genomics (2003) 269: 624–631
DOI 10.1007/s00438-003-0869-4

ORIGINAL PAPER

J. Macas · P. Neumann · D. Požárková

Zaba: a novel miniature transposable element present in genomes of legume plants

Received: 27 January 2003 / Accepted: 21 May 2003 / Published online: 30 July 2003
© Springer-Verlag 2003

Abstract A novel family of miniature transposable elements, named *Zaba*, was identified in pea (*Pisum sativum*) and subsequently also in other legume species using computer analysis of their DNA sequences. *Zaba* elements are 141–190 bp long, generate 10-bp target site duplications, and their terminal inverted repeats make up most of the sequence. *Zaba* elements thus resemble class 3 foldback transposons. The elements are only moderately repetitive in pea (tens to hundreds copies per haploid genome), but they are present in up to thousands of copies in the genomes of several *Medicago* and *Vicia* species. More detailed analysis of the elements from pea, including isolation of new sequences from a genomic library, revealed that a fraction of these elements are truncated, and that their last transposition probably did not occur recently. A search for *Zaba* sequences in EST databases showed that at least some elements are transcribed, most probably due to their association with genic regions.

Keywords Miniature transposable elements · Repetitive DNA · Plant genome · Computer analysis · Pea (*Pisum sativum*)

Introduction

Transposable elements are defined as discrete segments of DNA that are able to move within genomes, either directly (DNA transposons) or via RNA intermediates (retroelements). While the DNA transposons move using an excision and reintegration-based (cut-and-paste) mechanism and are only moderately repeated in the genome, the retroelements accumulate to high copy numbers due to their replicative mode of transposition (Bennetzen 2000). In higher plants, both classes are present and include a number of diverse families (Grandbastien 1992; Feschotte et al. 2002a). Relatively recently, a novel group of elements called MITEs (miniature inverted-repeat transposable elements) was described that is characterized by its small size (usually < 500 bp), lack of coding capacity, and the presence of terminal inverted repeats (TIRs) at both ends. First discovered in maize (Bureau and Wessler 1992), numerous families differing in their nucleotide sequences were later identified in both monocot and dicot plants, and also in animals (for review, see Feschotte et al. 2002a, 2002b). In spite of their diversity, most MITEs possess conserved features, such as the sequences of their TIRs and target site duplications (TSDs), that presumably reflect their mobilization by certain transposases. It is thus assumed that these elements represent non-autonomous deletion derivatives of longer DNA transposons (also called 'master elements') that encode transposases. Based on their association with established superfamilies of transposases, most MITEs can be assigned to one of two basic groups, Tc1/*Mariner* and *PIF*/*Harbinger* (Feschotte et al. 2002b). Some MITEs have accumulated to high copy numbers in plant genomes (Bureau and Wessler 1992; Mao et al. 2000; Zhang et al. 2000), which is unusual for DNA transposons and suggests the existence of an additional amplification mechanism acting on these elements (Río et al. 1996; Izstvák et al. 1999).

While most known MITE families can be classified as Tc1/*Mariner* or *PIF*/*Harbinger* -like, a number of miniature transposable elements could not be assigned to these groups. These include MITEs derived from unknown master elements (Feschotte et al. 2002b) and a group of type 3 foldback transposons that are

Electronic Supplementary Material Supplementary material is available for this article if you access the article at http://dx.doi.org/10.1007/s00438-003-0869-4. A link in the frame on the left on that page takes you directly to the supplementary material.

Communicated by M.-A. Grandbastien

J. Macas (✉) · P. Neumann · D. Požárková
Laboratory of Molecular Cytogenetics,
Institute of Plant Molecular Biology,
Branišovská 31, 37005 České Budějovice, Czech Republic
E-mail: macas@umbr.cas.cz
Tel.: +420 38-7775516
Fax: +420 38-5310356

significantly reduced in size compared to their corresponding full-length master elements (Rebatchouk and Narita 1997; Ade and Belzile 1999). The full-length (or type 1) foldback elements are composed of a central segment surrounded by long TIRs containing distinct outer and inner domains (Rebatchouk and Narita 1997). The type 3 elements contain only the inner domains of the TIRs and the short central segment, and thus resemble some MITEs such as *Bigfoot* (Charrier et al. 1999) whose inverted repeats also make up most of the element.

In the course of a project aimed at the characterization of repetitive DNA sequences in pea (*Pisum sativum* L.) we attempted to identify miniature transposable elements in this species. Although relatively detailed analyses of pea repetitive sequences have been performed using molecular techniques (Neumann et al. 2001), no such elements have been identified so far, except for two copies of the Tc1/*Mariner* -like MITE *Stowaway*, which were found using computer analysis of DNA sequences (Bureau and Wessler 1994). Thus, we used the same approach to search for additional miniature transposons in the growing collection of pea sequence data available in public databases. Here we describe the identification of a novel family of these elements, and its characterization with respect to sequence composition, abundance, and presence in other plant species.

Materials and methods

Computer analysis

A program for the identification of potential MITE elements was written in Perl, using BioPerl modules. This was used to search for inverted repeats of at least 12 bp (allowing for one mismatch) separated by up to 400 bp of any sequence. These search parameters were chosen based on the sequence features of known elements (Feschotte et al. 2002a, 2002b) so that most of them could be efficiently detected while ensuring that the computational demands of the analysis could be met by a Pentium III-based desktop computer. The search was performed in all genomic sequences of pea available from the GenBank database in November 2001. Potential elements and their surrounding regions were then tested for homology to other plant sequences using BLAST (Altschul et al. 1997), FASTA (Pearson and Lipman 1988), and Dotter (Sonnhammer and Durbin 1995). Homology searches in new GenBank entries using BLAST and FASTA were periodically repeated until December 2002. Further analysis of the elements was performed at the Biology WorkBench website (http://workbench.sdsc.edu/). The DNA folding calculations were done for a folding temperature of 37°C and an ionic strength of 1.0 M Na^+ using the mfold program implemented at http://www.bioinfo.rpi.edu/applications/mfold/old/dna/. This server uses free energies determined by SantaLucia (1998) and the salt correction established by J. SantaLucia Jr., M. Zuker, A. Bommarito and R. J. Irani (unpublished results).

Plant material and genomic DNA isolation

Seeds of plant species used in this study were obtained from the Plant Breeding Station at Borsov, Czech Republic (*P. sativum* L. cv. Carrera), the Western Regional Plant Introduction Station, Pullman, Wash., USA (*P. sativum* ssp. *sativum* (Accession No. PI116056), *P. sativum* ssp. *asiaticum* Govor. (W615036), *P. sativum* var. *arvense* (L.) Poir. (PI268480), *P. fulvum* Sm. (PI560063), *P. sativum* ssp. *elatius* (Steven ex M. Bieb.) Meikle (W615043), *P. sativum* var. *elatius* (Steven ex M. Bieb.) Meikle (W616220), *P. sativum* ssp. *abyssinicum* (A. Braun) Govor. (PI358610), *P. sativum* ssp. *transcaucasicum* Govor. (W612704), *P. sativum* var. *pumilio* Meikle (PI560068)), from the Research Institute for Fodder Plants, Troubsko, Czech Republic (*Medicago sativa* L. cv. Jarka, *M. truncatula* Gaertn.), Agritec, Sumperk, Czech Republic (*Vicia narbonensis* L., *V. sativa* L., *Phaseolus vulgaris* L., *Glycine max* (L.) Merr., *Lupinus angustifolius* L., *Cicer arietinum* L.), the Institut für Kulturpflanzenforschung (IPK), Gatersleben, Germany (*Vigna unguiculata* (L.) Walp.), and Dr. M. Vavak, Horna Streda, Slovakia (*Vicia faba* L. cv. Inovec). Total genomic DNA was extracted from leaves pooled from 4–5 plants as described by Dellaporta et al. (1983). All DNA concentration measurements were done using the PicoGreen dye (Molecular Probes) according to the manufacturer's recommendations.

Dot-blot and genomic library hybridizations

To estimate the copy numbers of *Zaba* elements in the genomes of individual species, serial dilutions of their genomic DNAs were quantitatively dot-blotted onto Hybond N+ membrane (Amersham) together with a cloned fragment of the pea vicilin gene containing a *Zaba-Ps1* element as a hybridization standard. The quantity of spotted genomic DNA was corrected for differences in genome size between species (equivalents of 10^3–10^5 copies of haploid genomes were compared to 10^6–5×10^9 copies of the hybridization standard). Dot-blots were hybridized with a 126-bp probe prepared by PCR amplification of part of the *Zaba-Ps1* sequence using the primers PSVICIL-X (5′-AATAGGTCC CACATAAAGTG-3′) and PSVICIL-R2 (5′-CAAAACAAATATGTAAAGTAATAAGAG-3′). The probe was labeled with alkaline phosphatase and hybridized using the AlkPhos Direct Kit (Amersham). Probe detection was performed using CDP Star substrate (Amersham) and the signals were captured on X-ray film. The hybridization and washing temperatures were varied from 45°C to 55°C to achieve moderate or high stringency.

For library screening, a pea genomic DNA library cloned into Lambda DASH II (purchased from Stratagene) was plated at high density, and about 100,000 clones were screened with the *Zaba-Ps1* probe using plaque lift hybridization. Purification of selected positive clones was done by several rounds of secondary screening using diluted phages recovered from the plates. Purified clones were used for restriction enzyme analysis and subcloning of fragments suitable for sequencing using standard procedures (Ausubel et al. 1991).

PCR amplification and cloning

Amplification of genomic loci containing *Zaba* elements was performed using 20 ng of genomic DNA as a template in 30-μl reactions composed of 1x PCR buffer, 0.2 mM dNTPs, 1.5 mM $MgCl_2$, 0.2 μM primers, and 1.2 U of Platinum *Taq* polymerase (Invitrogen). Primers for amplification of the locus carrying *Zaba-Ps1* (5′-TGCCGATTACATCCTTGTTG-3′and 5′-TCGGGTTTCAACACTGTGAG-3′) were derived from the *P. sativum* vicilin gene sequence (Accession No. X14076) and should yield a product of 402 bp. Primers for *Zaba-Ps2* (5′-CATTTGCAAGATGGTAACAATG-3′ and 5′-TCAAGCTCATCGAGAAACAG-3′) were based on the sequence of our clone c101e (Accession No. AY204902), and should amplify a region of 494 bp. PCR was performed for 30 cycles of 94°C for 1 min, 50°C (*Zaba-Ps2*) or 55°C (*Zaba-Ps1*) for 1 min, and 72°C for 1 min, preceded by an initial denaturation (94°C, 2 min) and followed by a final extension step (72°C, 10 min). The amplification products were analyzed on ethidium bromide-stained 2% agarose gels, and the presence of *Zaba* sequences in amplified fragments was investigated using

Southern hybridization with a partial *Zaba-Ps1* probe as described for dot-blots. Selected amplification products were cloned using the TOPO TA Cloning Kit (Invitrogen) and sequenced by the dideoxy chain termination method (Sanger et al. 1977) using the Big Dye Terminator Cycle Sequencing Kit and the ABI PRISM 310 Genetic Analyzer (Applied Biosystems).

Results

Identification and sequence analysis of *Zaba* elements

Computer analysis was used to search all the genomic DNA sequences from pea that are available in public databases for loci containing inverted repeats of at least 12 bp separated by up to 400 bp. These regions were used as queries in BLAST and FASTA searches of plant DNA sequences in order to identify those that share homology and are flanked by otherwise unrelated sequences. Based on this analysis, two groups of elements were identified. The first group included three copies of the *Stowaway* element, two of which had been described previously (Bureau and Wessler 1994). The third element was found at positions 1901–2175 in the intron of the CYP73A9 gene (Accession No. AF175275) and shared high homology with the other two elements, as well as with *Stowaway* MITEs from a variety of plant species (data not shown). The second group included a single inverted repeat sequence present in the first intron of a vicilin gene (X14076, positions 1963–2151). As this sequence did not show homology to any known mobile elements but displayed some of their characteristic features, it was investigated in more detail.

A comparison of the vicilin sequence X14076 with highly homologous genes from pea (vicilin J, X67428), *Vicia faba* and *Lens odemensis* revealed that only the first contains the potential element. Alignment of these vicilin sequences allowed precise determination of the element ends and revealed a duplication of 10 bp of the flanking sequence (Fig. 1A). This target site duplication (TSD), together with the presence of the insertion in only one of the four vicilin genes, provided strong evidence for its classification as a novel miniature transposable element. Moreover, similar sequences were found in otherwise non-homologous genomic regions in three other legume species, *Lotus japonicus* , *Medicago sativa* and *M. truncatula* (Table 1). Thus, it was concluded that they all represent members of the same family of mobile elements, which we named *Zaba*.

Multiple sequence alignments and analysis of adjacent sequences showed that *Zaba* elements are 141–190 bp long and generate 10-bp target-site duplications. The TSDs were clearly recognizable in seven out of the sixteen identified elements, and were always conserved in size but not in their nucleotide sequence (Fig. 1B). Mutual homology of individual elements ranged from 53% to 90% (72% on average) and was generally higher for those from the same species (data not shown). Compared to the average base composition of plant genomes (Barow and Meister 2001) the elements

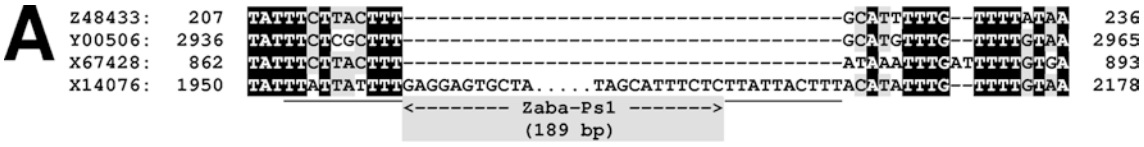
Fig. 1A, B Identification and multiple sequence alignment of *Zaba* elements. **A** Sequence comparison between vicilin sequences from *Lens odemensis* (Accession No. Z48433), *Vicia faba* (Y00506), and *Pisum sativum* (X67428 and X14076). GenBank accession numbers and nucleotide positions are indicated for each sequence. The position of *Zaba-Ps1* in X14076 is marked *below* the alignment and the target site duplication is *underlined*. **B** Multiple sequence alignment of identified *Zaba* elements. The element sequences are displayed in *upper case letters*, and completely or partially conserved nucleotides are indicated on *black* and *gray* backgrounds, respectively. Flanking sequences are in *lower case*, and distinguishable target site duplications are *underlined*. The "X" in the *Zaba-Mt10* sequence marks the position of a 130-bp insertion of heterologous sequence that was excluded from the alignment. Further information about each element is given in Table 1

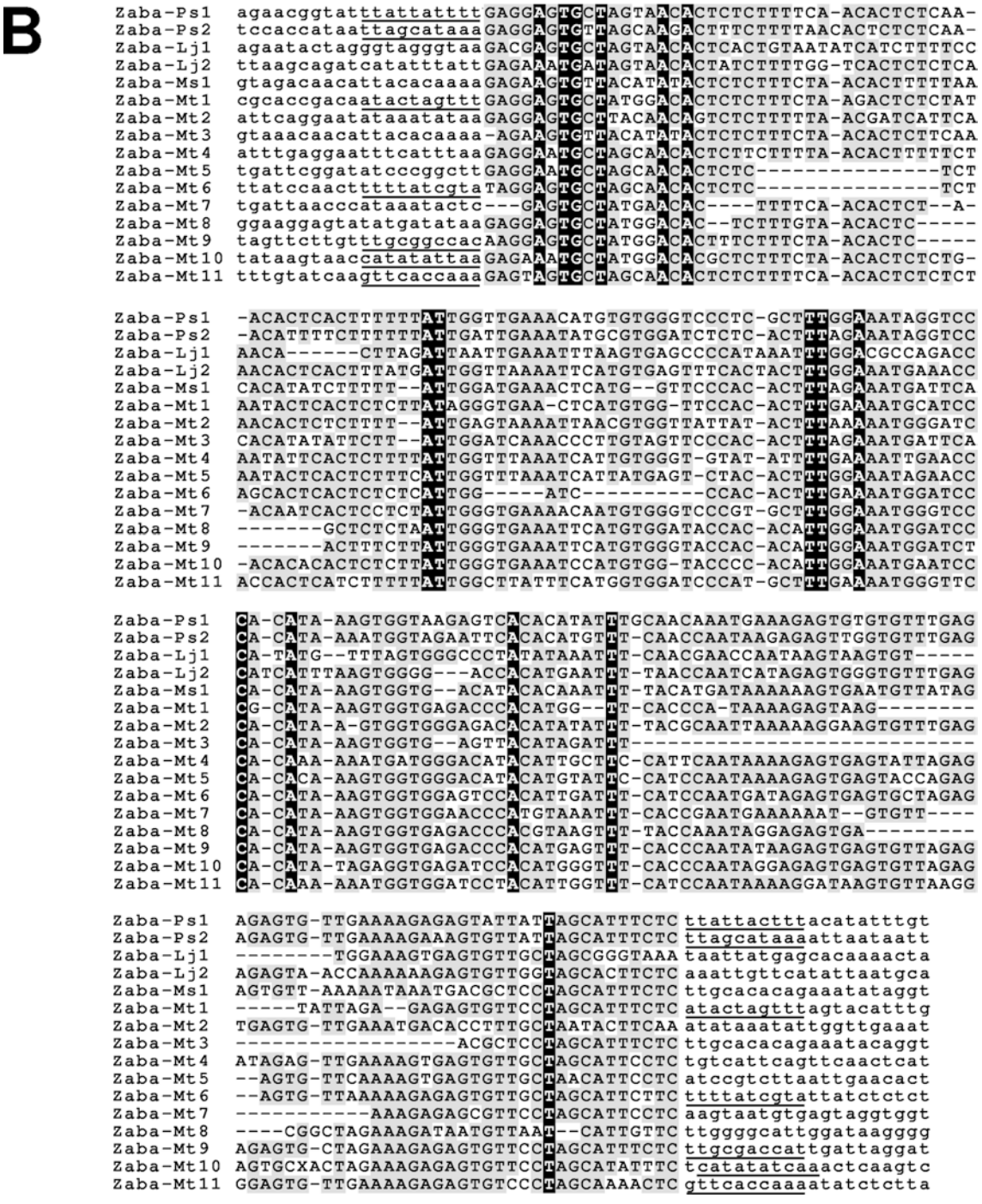
are not particularly rich in A/T (62% A/T), and contain terminal inverted repeats that account for most of their length. The DNA folding predictions suggested that most of the elements have the potential to form secondary structures due to their long TIRs (Fig. 2). The exceptions are *Zaba-Mt3*, lacking a large part of its right TIR, and *Zaba-Mt10*, which is disrupted by the insertion of a heterologous sequence that shows strong homology to the retroelement *Tms1* (Végh et al. 1990). The stability of the predicted hairpin structures in other elements (expressed as the free energy) ranges from −17.0 (*Zaba-Ms1*) to −59.7 kcal/mol (*Zaba-Lj2*), with the average of −37.0 kcal/mol, which reflects the degree of sequence divergence of their TIRs.

A FASTA search in EST databases revealed a number of plant transcripts containing *Zaba* sequences. The best homology was found to six ESTs from *M. truncatula* that gave scores (expectation values) lower than $1e^{-10}$. These transcripts contained truncated as well as full-length elements (Table 2). A total of 34 unique transcripts gave expectation values lower than $1e^{-5}$ and included sequences from *M. truncatula*, *G. max* and *L. japonicus* (not shown). Of the six ESTs listed in Table 2, two (BG588203 and BQ123642) could be assigned to putative genes based on their homology to annotated sequences from *Arabidopsis thaliana* (Accession No. NP_188239) and rice (Accession No. AC108884), respectively (data not shown). This is an indication that at least some *Zaba* elements are transcribed, most probably due to their association with gene sequences.

Species distribution and abundance

In addition to database searches, the species distribution of *Zaba* sequences was also examined experimentally using quantitative dot blot hybridization of genomic DNA from selected legumes and *A. thaliana*. As full-length *Zaba* sequences may form secondary structures (Fig. 2), the 126-bp hybridization probe was derived from the central and 3′ segments of *Zaba-Ps1* in order to avoid intrastrand self-annealing. The hybridizations were performed at various stringency conditions in order





to allow efficient detection of relatively heterogeneous *Zaba* sequences. The experiments using moderate stringency showed that the elements are most abundant in *M. sativa, M. truncatula, V. faba* and *V. narbonensis* (estimated copy numbers of 10^3–3×10^3 per haploid genome). They were also detected in *P. sativum* and *P. fulvum* (10^2–10^3 copies/1C), but were undetectable in other species (the detection limit of the assay was about 50–100 copies/1C) (Fig. 3). The use of high hybridization stringencies resulted in a proportional (about

Table 1 *Zaba* elements

Element	Length [bp]	Accession No.	Definition[a]	Position[b]	Notes
Zaba-Ps1	189	X14076	*Pisum sativum* vicilin gene	1963–2151 (intron 1)	–
Zaba-Ps2	189	AY204902	*P. sativum* genomic clone c101e	260–448	Sequence isolated in this study
Zaba-Lj1	173	Y12859	*Lotus japonicus* gln1, pge1 and krm genes	12893–13065 (r) (intron 7 of gln1)	–
Zaba-Lj2	190	AP004503	*L. japonicus* genomic DNA, chromosome 5, clone:LjT02M03, TM0034a	29056–29245	–
Zaba-Ms1	184	U35073	*Medicago sativa* nodulin (ENOD8) gene	469–652 (promoter)	Homology to AF463407
Zaba-Mt1	171	AF194421	*M. truncatula* homoglutathione synthetase GSHS2 (gshs2) gene, partial cds; and glutathione synthetase GSHS1 (gshs1) gene, complete cds	1044–1214 (r) (intron 4 of gshs2)	–
Zaba-Mt2	187	AC087771	Genomic sequence for *M. truncatula*, clone 8D15, complete sequence	37566–37752 (r)	–
Zaba-Mt3	141	AF463407	*M. truncatula* Enod8.3 (Enod8.3) gene, partial cds; and Enod8.2 (Enod8.2) and Enod8.1 (Enod8.1) genes, complete cds	9656–9796 (promoter of Enod8.1)	Homology to U35073
Zaba-Mt4	189	AC127429.9	*M. truncatula* clone mth2-9o20 (WDS)	95853–96041	–
Zaba-Mt5	173	AC121239.9	*M. truncatula* clone mth1-8p19 (WDS)	7422–7594 (r)	–
Zaba-Mt6	159	AC130802.6	*M. truncatula* clone mth2-28b21 (WDS)	112281–112439 (r)	–
Zaba-Mt7	163	AC124967.5	*M. truncatula* clone mth2-13h21 (WDS)	69652–69814	–
Zaba-Mt8	162	AC127429.9	*M. truncatula* clone mth2-9o20 (WDS)	95261–95422 (r)	–
Zaba-Mt9	178	AC121245.6	*M. truncatula* clone mth2-33g3 (WDS)	44175–44352 (r)	–
Zaba-Mt10	187	AC122161.8	*M. truncatula* clone mth2-11l3 (WDS)	95633–95789 +95920–95949	Contains aninsertion of heterologous sequence (95790–95919)
Zaba-Mt11	190	AC125478.7	*M. truncatula* clone mth2-31i19 (WDS)	90992–91181 (r)	–

[a]Definition of the sequence containing the element. Unfinished sequences are marked as WDS (working draft sequence)
[b]Position of the element within the sequence [(r) indicates that it is present in reverse orientation]

10-fold) decrease in the signal in all species (not shown). The copy number estimation obtained for the pea genome was also checked using hybridization of the *Zaba-Ps1* probe to a pea genomic library cloned into a phage vector. Seventeen positive clones were detected among about 100,000 plaques screened, which is equivalent to ~50 copies per haploid genome, provided that each clone contained only a single element. Thus, based on these assays, and taking the sequence heterogeneity of the elements into account, the abundance of *Zaba* in the pea genome is probably in the range of 50 to 500 copies/1C.

Characterization of *Zaba* elements in the pea genome

In order to obtain more *Zaba* sequences from the pea genome, four randomly selected phage clones that hybridized with the *Zaba-Ps1* probe were purified and subjected to restriction analysis, and the fragments that hybridized with the probe were subcloned. Analysis of the resulting sequences (875–2140 bp in length) showed that three of them contained only truncated elements. The remaining clone included a full-length element designated *Zaba-Ps2* (Table 1) that was 189 bp long and had a 10-bp TSD (Fig. 1B).

To investigate possible insertional polymorphism of the two full-length elements from pea (*Zaba-Ps1* and *Zaba-Ps2*), primers derived from their surrounding sequences were used to amplify corresponding loci from nine accessions of *P. sativum* and from *P. fulvum* (Fig. 4). All taxa yielded products of the size expected for the sequence containing the element; however, one accession (*P. sativum* ssp. *transcaucasicum*, Fig. 4, lane 8) produced additional fragments of different sizes. These fragments, as well as PCR products of the expected size from *P. sativum* ssp. *transcaucasicum* and *P. sativum* cv. Carrera (Fig. 4, lane 1) as controls, were cloned and sequenced. Sequence comparisons revealed that the additional, slightly longer, fragment generated

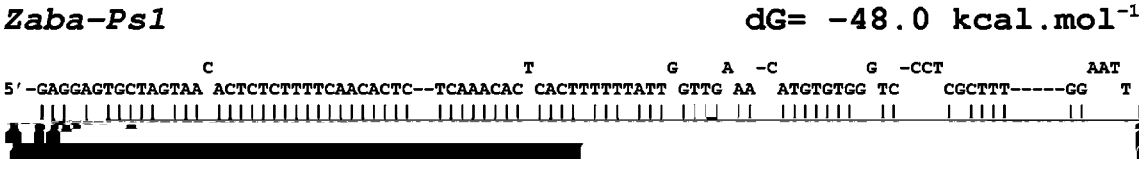


Fig. 2 Predicted DNA secondary structure of *Zaba-Ps1*. The free energy (ΔG) calculated for this sequence is given *above* the structure

Table 2 ESTs containing *Zaba* elements

Accession	Length (bp) [a]	Homology[b]		Element[c]
		Position	Score	
BQ139424	661	499–661	$2.3e^{-26}$	3′ truncated
BG588203	658	327–507	$9.2e^{-21}$	Full-length
BG451390	676	80–274	$5.5e^{-17}$	Full-length
BQ123048	654	112–244	$6.3e^{-12}$	Not truncated, but shows only partial homology
BQ123642	660	1–131	$1.2e^{-11}$	5′ truncated
BE941999	564	445–564	$2.7e^{-11}$	3′ truncated

[a]Total length of EST sequence
[b]Homology is given by its position within EST sequence and FASTA score (expectation value)
[c]Indicates whether the element sequence is truncated because it occurs at either end of the EST

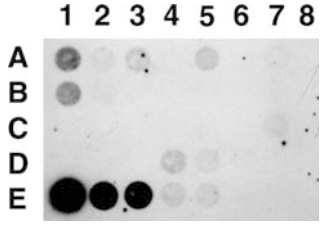


Fig. 3 Abundance of *Zaba* elements in selected plant species. A quantitative dot-blot hybridization of a partial *Zaba-Ps1* probe to 10^5 and 10^4 haploid genome equivalents of *Medicago sativa* (A1, A2), *M. truncatula* (A3, A4), *Vicia faba* (A5, A6), *V. sativa* (A7, A8), *V. narbonensis* (B1, B2), *Cicer arietinum* (B3, B4), *Glycine max* (B5, B6), *Lupinus angustifolius* (B7, B8), *Phaseolus vulgaris* (C1, C2), *Vigna unguiculata* (C3, C4), *Arabidopsis thaliana* (C5, C6), and *Pisum fulvum* (C7, C8) is shown. The dots at positions D4–D8 contain serial dilutions of *P. sativum* DNA (10^5, 5×10^4, 10^4, 5×10^3, 10^3) and positions E1–E8 contain hybridization standards (5×10^9, 10^9, 5×10^8, 10^8, 5×10^7, 10^7, 5×10^6, and 10^6 copies of a plasmid carrying the *Zaba-Ps1* element)

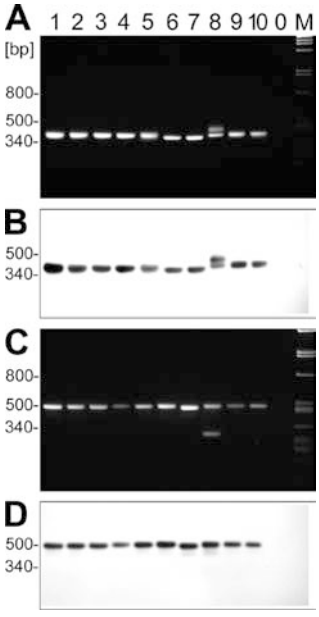


Fig. 4A–D Amplification of *Zaba-Ps1* and *Zaba-Ps2* loci from various *Pisum* accessions. Primer pairs derived from sequences surrounding the *Zaba-Ps1* (**A**) and *Zaba-Ps2* (**C**) insertion sites were used to amplify corresponding loci from *P. sativum* cv. Carrera (lane 1), *P. sativum* ssp. *sativum* (lane 2), *P. sativum* ssp. *elatius* (lane 3), *P. sativum* var. *elatius* (lane 4), *P. sativum* var. *arvense* (lane 5), *P. sativum* var. *pumilio* (lane 6), *P. sativum* ssp. *abyssinicum* (lane 7), *P. sativum* ssp. *transcaucasicum* (lane 8), *P. sativum* ssp. *asiaticum* (lane 9), and *P. fulvum* (lane 10). The amplification products were resolved on ethidium bromide-stained 2% agarose gels. Lane 0 contains PCR reaction with no template (negative control), and lane M contains lambda DNA digested with *Pst* I. Southern blots of the gels displayed in panels **A** and **C** hybridized with a partial *Zaba-Ps1* probe are shown in panels **B** and **D**, respectively

in amplifications of the *Zaba-Ps1* locus (Fig. 4A, lane 8) originated from an allele containing a 54-bp duplication within the element sequence itself (see Fig. S1 of the Electronic Supplementary Material for the sequences). On the other hand, an additional fragment generated in amplifications of the *Zaba-Ps2* locus from *P. sativum* ssp. *transcaucasicum* DNA was shorter than the expected product (Fig. 4B, lane 8). It was found to be almost identical to the longer fragment, but it lacks the whole *Zaba* element sequence (Electronic Supplementary Material, Fig. S2). As the target site duplication was also missing from the shorter fragment it is most likely that it represents the original sequence where element integration did not occur.

Discussion

As miniature transposable elements are often divergent in their nucleotide sequences, even if they belong to the same family, their classification is mainly based on other features, such as target site preference, target site duplications, and conserved size and/or sequence motifs in the terminal inverted repeats (Feschotte et al. 2002a, 2002b). Based on these characteristics, *Zaba* elements cannot be assigned to any of the currently recognized superfamilies of MITEs. They do not show any target sequence preference, generate 10-bp TSDs (unusually long for MITEs), and their TIRs—which extend over almost the whole sequence—do not show significant homology to other elements. With respect to the MITEs described so far, some of these features resemble those of *Bigfoot* elements of *Medicago* (Charrier et al. 1999); however, *Bigfoot* generates only 9-bp TSDs and is relatively A/T rich (about 72%, in contrast to 62% for *Zaba*). Moreover, no sequence homology was detected between *Bigfoot* and *Zaba* elements (not shown).

Based on the data obtained so far, *Zaba* elements appear to be specific for legumes. However, their abundance in individual species varies over three orders of magnitude, reaching up to several thousands of copies per haploid genome in some *Medicago* and *Vicia* species. Although the number of annotated sequence entries found to contain *Zaba* elements is limited, it suggests

that at least a fraction of these elements is associated with coding sequences (Table 1). This is supported by the identification of *Zaba* sequences in EST databases, which is probably a consequence of their location in transcribed regions. As accumulation to high copy numbers and association with coding sequences are features that have been described for several MITE families (Bureau and Wessler 1992, 1994; Mao et al. 2000), it is possible that *Zaba* elements and MITEs are maintained and propagated in the genome by similar mechanisms (Río et al. 1996; Izstvák et al. 1999). On the other hand, it is still possible that *Zaba* sequences are not related to MITEs, but represent type 3 foldback elements lacking some parts of the full-length master transposons. This would be in accordance with the presence of the long TIRs in *Zaba*, as well as with the 10-bp TSDs which have been described for some plant foldback transposons. Moreover, SoFT foldback elements of *Solanaceae* can also attain high copy numbers (Rebatchouk and Narita 1997). Thus, a reliable classification of *Zaba* is not possible at present, and has to be postponed until more genomic sequence data are available for the identification of potential master elements from which *Zaba* could possibly be derived.

In addition to the identification and general characterization of *Zaba* as a new family of mobile elements, this study focused on a more detailed analysis of corresponding sequences present in the pea genome. The copy number estimation using two different approaches showed that pea contains tens to hundreds of *Zaba* sequences. However, some of these copies are truncated, as revealed by sequencing of clones retrieved from the genomic DNA library. The occurrence of these decayed elements suggests that amplification of *Zaba* in the pea genome is not recent. This hypothesis is supported by a PCR survey of ten *Pisum* accessions for the presence of *Zaba-Ps1* and *Zaba-Ps2*, which confirmed their occurrence at corresponding loci in all taxa tested. This can most easily be explained by their transposition into these loci early in the evolutionary history of the genus. However, the finding of an additional allele lacking *Zaba-Ps2* in the sample of *P. sativum* ssp. *transcaucasicum* DNA appears to contradict this explanation. A further transposition would result in the removal of the element from its original location and could be limited to this taxon, which probably represents a distinct evolutionary lineage within the genus (Pearce et al. 2000). However, such an event should leave a footprint corresponding to the target site duplication. As *Zaba-Ps2* was flanked by clearly distinguishable TSDs which were not found in the sequence lacking the element (Fig. 1B, and Fig. S1 Electronic Supplementary Material) it is likely that the element never inserted into this allele. On the other hand, as the nature of the *Zaba-Ps2* flanking sequences is not known, we cannot rule out the possibility that the fragment lacking the element represents an additional locus of a multicopy sequence rather than another allele of the same locus.

Acknowledgements We thank Ms. H. Štěpančíková and Ms. O. Šonková for excellent technical assistance, and Ms. S. Rafelski for assistance in the preparation of the manuscript. We are also grateful to the following institutions and individuals for providing the seeds of plant species used in this study: the Western Regional Plant Introduction Station, Pullman, Wash., USA; the Plant Breeding Station at Borsov, Czech Republic; the Research Institute for Fodder Plants, Troubsko, Czech Republic; AGRITEC, Sumperk, Czech Republic; IPK Gatersleben, Germany; and Dr. M. Vavak, Horna Streda, Slovakia. This work was supported by Grant No. 521/00/0655 from the Grant Agency of the Czech Republic. The experiments have been carried out in compliance with the current laws governing genetic experimentation in the Czech Republic.

References

Ade J, Belzile FJ (1999) *Hairpin* elements, the first family of foldback transposons (FTs) in *Arabidopsis thaliana*. Plant J 19:591–597

Altschul SF, Madden TL, Schäffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res 25:3389–3402

Ausubel FM, Brent R, Kingston RE, Moore DD, Seidman JG, Smith JA, Struhl K (1991) Current protocols in molecular biology. Wiley, New York

Barow M, Meister A (2001) Lack of correlation between AT frequency and genome size in higher plants and the effect of nonrandomness of base sequences on dye binding. Cytometry 47:1–7

Bennetzen JL (2000) Transposable element contributions to plant gene and genome evolution. Plant Mol Biol 42:251–269

Bureau TE, Wessler SR (1992) *Tourist*: a large family of small inverted repeat elements frequently associated with maize genes. Plant Cell 4:1283–1294

Bureau TE, Wessler SR (1994) *Stowaway*: a new family of inverted repeat elements associated with the genes of both monocotyledonous and dicotyledonous plants. Plant Cell 6:907–916

Charrier B, Foucher F, Kondorosi E, d'Aubenton-Carafa Y, Thermes C, Kondorosi A, Ratet P (1999) *Bigfoot*: a new family of MITE elements characterized from the *Medicago* genus. Plant J 18:431–441

Dellaporta SL, Wood J, Hicks JB (1983) A plant DNA minipreparation: version II. Plant Mol Biol Rep 1:19–21

Feschotte C, Jiang N, Wessler SR (2002a) Plant transposable elements: where genetics meets genomics. Nature Rev Genet 3:329–341

Feschotte C, Zhang X, Wessler SR (2002b) Miniature inverted-repeat transposable elements and their relationships to established DNA transposons. In: Craig NL, Craigie R, Gellert M, Lambowitz AM (eds) Mobile DNA II. ASM Press, Washington, D.C., pp 1147–1157

Grandbastien M-A (1992) Retroelements in higher plants. Trends Genet 8:103–108

Izstvák Z, Ivics Z, Shimoda N, Mohn D, Okamoto H, Hackett PB (1999) Short inverted-repeat transposable elements in teleost fish and implications for a mechanism of their amplification. J Mol Evol 48:13–21

Mao L, Wood TC, Yu YS, Budiman MA, Tomkins J, Woo SS, Sasinowski M, Presting G, Frisch D, Goff S, Dean RA, Wing RA (2000) Rice transposable elements: a survey of 73,000 sequence-tagged-connectors. Genome Res 10:982–990

Neumann P, Nouzová M, Macas J (2001) Molecular and cytogenetic analysis of repetitive DNA in pea (*Pisum sativum* L.). Genome 44:716–728

Pearce SR, Knox M, Ellis THN, Flavell AJ, Kumar A (2000) Pea Ty1- *copia* group of retrotransposons: transcriptional activity and use as markers to study genetic diversity in *Pisum*. Mol Gen Genet 236:898–907

Pearson WR, Lipman DJ (1988) Improved tools for biological sequence comparison. Proc Natl Acad Sci USA 85:2444–2448

Rebatchouk D, Narita JO (1997) Foldback transposable elements in plants. Plant Mol Biol 34:831–835

Río A, Puigdomenech P, Casacuberta JM (1996) *Mrs*, a new family of *Tourist* transposable elements. Plant Mol Biol 32:1221–1226

Sanger F, Nicklen D, Coulson AR (1977) DNA sequencing with chain terminating inhibitors. Proc Natl Acad Sci USA 74:5463–5467

SantaLucia J Jr (1998) A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. Proc Natl Acad Sci USA 95:1460–1465

Sonnhammer ELL, Durbin R (1995) A dot-matrix program with dynamic threshold control suited for genomic DNA and protein sequence analysis. Gene 167:GC1–10

Végh Z, Vincze E, Kadirov R, Tóth G, Kiss GB (1990) The nucleotide sequence of a nodule-specific gene, *Nms-25* of *Medicago sativa*: its primary evolution via exon-shuffling and retrotransposon-mediated DNA rearrangements. Plant Mol Biol 15:295–306

Zhang Q, Arbuckle J, Wessler SR (2000) Recent, extensive, and preferential insertion of members of the miniature inverted-repeat transposable element family *Heartbreaker* into genic regions of maize. Proc Natl Acad Sci USA 97:1160–1165